

Variational Empirical-Bayes Poisson NMF and Effective-Rank Selection

1 Purpose and main qualification

Let $Y \in \mathbb{N}^{N \times M}$ be a count matrix. We begin with a deliberately large candidate rank $K_{\max}$ and investigate whether factor-specific Gamma priors on the loading columns can suppress unnecessary factors. The proposed model uses

$$L_{ik} \mid \alpha_k^0, \beta_k^0 \stackrel{\text{ind.}}{\sim} \text{Gam}(\alpha_k^0, \beta_k^0),$$

where the Gamma distribution is parameterized by shape and rate. A mean-field approximation is used for L , and (α_k^0, β_k^0) are estimated by variational empirical Bayes (VEB).

This procedure keeps $K_{\max}$ fixed and may estimate a smaller *effective rank*. It does not put a posterior distribution directly on K . An unused factor has a VEB fixed-point direction in which its prior mean α_k^0/β_k^0 approaches zero. Nevertheless, unconstrained VEB is not guaranteed to remove redundant factors: two similar factors may divide the signal and both retain positive prior means. Section 7 explains this qualification precisely.

2 Poisson NMF model

The observation model is

$$Y_{im} \mid L, F \stackrel{\text{ind.}}{\sim} \text{Poi}\left(\sum_{k=1}^{K_{\max}} L_{ik} F_{km}\right), \quad (1)$$

$$L_{ik} \mid \alpha_k^0, \beta_k^0 \stackrel{\text{ind.}}{\sim} \text{Gam}(\alpha_k^0, \beta_k^0). \quad (2)$$

We impose

$$F_{km} \geq 0, \quad \sum_{m=1}^M F_{km} = 1, \quad k = 1, \dots, K_{\max}. \quad (3)$$

Under this normalization, L_{ik} is the expected total count contributed by factor k to observation i . Consequently, the prior mean α_k^0/β_k^0 has a directly interpretable scale.

The model can be initialized with vanilla Poisson NMF. If the initial factorization is $Y \approx \tilde{L}\tilde{F}$, define

$$F_{km}^{(0)} = \frac{\tilde{F}_{km}}{\sum_{m'=1}^M \tilde{F}_{km'}}, \quad L_{ik}^{(0)} = \tilde{L}_{ik} \sum_{m=1}^M \tilde{F}_{km}. \quad (4)$$

This removes the ordinary NMF scale ambiguity without changing the fitted mean.

3 Poisson allocation augmentation

Introduce latent component-specific counts

$$Z_{imk} \mid L, F \sim \text{Poi}(L_{ik}F_{km}), \quad Y_{im} = \sum_{k=1}^{K_{\max}} Z_{imk}. \quad (5)$$

Conditional on Y_{im} , their variational distribution is multinomial:

$$q(Z_{im1}, \dots, Z_{imK_{\max}} \mid Y_{im}) = \text{Multinomial}(Y_{im}; \xi_{im1}, \dots, \xi_{imK_{\max}}), \quad (6)$$

where $\xi_{imk} \geq 0$ and $\sum_k \xi_{imk} = 1$.

We use the mean-field family

$$q(L, Z) = \left\{ \prod_{i=1}^N \prod_{k=1}^{K_{\max}} \text{Gam}(L_{ik}; \alpha_{ik}, \beta_{ik}) \right\} \left\{ \prod_{i=1}^N \prod_{m=1}^M q(Z_{im} \mid Y_{im}) \right\}. \quad (7)$$

4 Variational lower bound

Up to the constants $-\sum_{i,m} \log(Y_{im}!)$, the augmented variational lower bound is

$$\begin{aligned} \mathcal{L} = & \sum_{i=1}^N \sum_{m=1}^M \sum_{k=1}^{K_{\max}} Y_{im} \xi_{imk} \{ \psi(\alpha_{ik}) - \log \beta_{ik} + \log F_{km} - \log \xi_{imk} \} \\ & - \sum_{i=1}^N \sum_{k=1}^{K_{\max}} \frac{\alpha_{ik}}{\beta_{ik}} \sum_{m=1}^M F_{km} \\ & - \sum_{i=1}^N \sum_{k=1}^{K_{\max}} \text{KL}\{\text{Gam}(\alpha_{ik}, \beta_{ik}) \parallel \text{Gam}(\alpha_k^0, \beta_k^0)\}. \end{aligned} \quad (8)$$

We use the conventions $0 \log 0 = 0$ and $\log F_{km} = -\infty$ when $F_{km} = 0$. In computation, the log is protected by a small numerical constant, but the factor update itself may preserve exact zeros if desired.

For Gamma distributions parameterized by shape and rate,

$$\begin{aligned} & \text{KL}\{\text{Gam}(\alpha, \beta) \parallel \text{Gam}(\alpha^0, \beta^0)\} \\ & = (\alpha - \alpha^0) \psi(\alpha) + \alpha^0 \log \frac{\beta}{\beta^0} - \log \Gamma(\alpha) + \log \Gamma(\alpha^0) - \alpha + \frac{\alpha \beta^0}{\beta}. \end{aligned} \quad (9)$$

5 Coordinate updates

5.1 Latent count allocations

For every i, m, k ,

$$\xi_{imk} = \frac{F_{km} \exp\{\psi(\alpha_{ik}) - \log \beta_{ik}\}}{\sum_{h=1}^{K_{\max}} F_{hm} \exp\{\psi(\alpha_{ih}) - \log \beta_{ih}\}}. \quad (10)$$

Only entries with $Y_{im} > 0$ contribute to subsequent sufficient statistics, so the implementation need not store ξ_{imk} for zero observations.

5.2 Loading posterior

Conjugacy gives

$$\alpha_{ik} \leftarrow \alpha_k^0 + \sum_{m=1}^M Y_{im} \xi_{imk}, \quad (11)$$

$$\beta_{ik} \leftarrow \beta_k^0 + \sum_{m=1}^M F_{km}. \quad (12)$$

Because of Eq. (3),

$$\beta_{ik} = \beta_k^0 + 1. \quad (13)$$

5.3 Factor dictionary

Let

$$C_{km} = \sum_{i=1}^N Y_{im} \xi_{imk}. \quad (14)$$

Maximizing the lower bound over a row-normalized F gives

$$F_{km} \leftarrow \frac{C_{km}}{\sum_{m'=1}^M C_{km'}}. \quad (15)$$

If $\sum_m C_{km}$ is numerically zero, the factor is provisionally inactive; its previous row may be retained until the final pruning decision. Damping the update can improve stability when factors are nearly redundant.

5.4 Variational empirical-Bayes update

For each candidate factor k , define the posterior averages

$$\bar{\lambda}_k = \frac{1}{N} \sum_{i=1}^N \frac{\alpha_{ik}}{\beta_{ik}}, \quad (16)$$

$$\overline{\log \lambda}_k = \frac{1}{N} \sum_{i=1}^N \{\psi(\alpha_{ik}) - \log \beta_{ik}\}. \quad (17)$$

Maximizing

$$\sum_{i=1}^N \mathbb{E}_q [\log \text{Gam}(L_{ik}; \alpha_k^0, \beta_k^0)]$$

over (α_k^0, β_k^0) gives

$$\beta_k^0 \leftarrow \frac{\alpha_k^0}{\bar{\lambda}_k}, \quad (18)$$

$$\log \alpha_k^0 - \psi(\alpha_k^0) = \log \bar{\lambda}_k - \overline{\log \lambda}_k. \quad (19)$$

Equation (19) can be solved by Newton's method. If

$$g(\alpha) = \log \alpha - \psi(\alpha) - \log \bar{\lambda}_k + \overline{\log \lambda}_k,$$

then

$$\alpha \leftarrow \alpha - \frac{g(\alpha)}{1/\alpha - \psi_1(\alpha)}, \quad (20)$$

where ψ_1 is the trigamma function. Numerical lower and upper bounds on α_k^0 and damping of both hyperparameter updates are recommended.

6 Inference algorithm

Algorithm 1 VEB Poisson NMF initialized by vanilla NMF

Require: Count matrix Y , candidate rank $K_{\max}$, inner iterations R , outer iterations T , convergence tolerance δ , and numerical constant ϵ .

- 1: Fit rank- $K_{\max}$ vanilla Poisson NMF to obtain $(\tilde{L}, \tilde{F})$.
 - 2: Normalize the factor rows and rescale the loading columns using Eq. (4); call the results $(L^{(0)}, F^{(0)})$.
 - 3: Initialize $\xi_{imk} \propto L_{ik}^{(0)} F_{km}^{(0)}$.
 - 4: Initialize positive (α_k^0, β_k^0) from the columns of $L^{(0)}$, and initialize $(\alpha_{ik}, \beta_{ik})$ using Eqs. (11)–(12).
 - 5: **for** $t = 1, \dots, T$ **do**
 - 6: **for** $r = 1, \dots, R$ **do**
 - 7: Update ξ using Eq. (10), retaining only the sufficient statistics needed for nonzero Y_{im} .
 - 8: Update $(\alpha_{ik}, \beta_{ik})$ using Eqs. (11)–(12).
 - 9: **end for**
 - 10: Update F using Eq. (15); optionally damp and then row-normalize.
 - 11: Update (α_k^0, β_k^0) independently for every k using Eqs. (18)–(20); optionally damp.
 - 12: Evaluate the lower bound in Eq. (8).
 - 13: **if** the relative lower-bound improvement is less than δ **then**
 - 14: **break**
 - 15: **end if**
 - 16: **end for**
 - 17: Recompute ξ and $q(L)$ using the final F and prior parameters.
 - 18: Calculate the relevance summaries in Eqs. (21) and (22), and prune only after convergence.
 - 19: **return** F , $(\alpha_{ik}, \beta_{ik})$, (α_k^0, β_k^0) , the lower-bound history, and $\hat{K}_{\text{eff}}$.
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7 Can this procedure learn the rank?

The natural VEB relevance summary is

$$r_k = \frac{\alpha_k^0}{\beta_k^0}. \quad (21)$$

At an interior VEB optimum, $r_k = \bar{\lambda}_k$. A complementary data-allocation summary is

$$p_k = \frac{\sum_{i=1}^N \sum_{m=1}^M Y_{im} \xi_{imk}}{\sum_{i=1}^N \sum_{m=1}^M Y_{im}}. \quad (22)$$

For an exploratory implementation, one may report

$$\hat{K}_{\text{eff}}(\tau) = \# \left\{ k : \frac{r_k}{\sum_{h=1}^{K_{\max}} r_h} \geq \tau \text{ and } p_k \geq \tau \right\}. \quad (23)$$

This is a thresholded effective rank, not a posterior estimate of K .

There is a real shrinkage mechanism. Suppose factor k receives no counts, so that $\sum_m Y_{im}\xi_{imk} = 0$ for every i . Then

$$\alpha_{ik} = \alpha_k^0, \quad \beta_{ik} = \beta_k^0 + 1.$$

The next undamped VEB update leaves the shape unchanged and maps

$$\beta_k^0 \leftarrow \beta_k^0 + 1.$$

Repeated updates therefore drive $r_k = \alpha_k^0/\beta_k^0$ toward zero. In this sense, an empty component can be automatically suppressed.

The difficulty is that an overfitted component need not remain empty. The factor update may adapt it to sampling noise, after which it receives counts and escapes shrinkage. More fundamentally, suppose two factor rows are identical, $F_1. = F_2.$, and their loading priors have a common rate:

$$L_{i1} \sim \text{Gam}(\alpha_1^0, \beta^0), \quad L_{i2} \sim \text{Gam}(\alpha_2^0, \beta^0).$$

Then

$$L_{i1} + L_{i2} \sim \text{Gam}(\alpha_1^0 + \alpha_2^0, \beta^0). \quad (24)$$

The likelihood cannot distinguish this two-factor representation from a single identical factor having shape $\alpha_1^0 + \alpha_2^0$. Consequently, VEB may have both a pruned solution and a signal-splitting solution, with the selected solution depending on initialization and optimization.

8 Relationship to conventional ARD

In the ARD-NMF model of [1], a common relevance scale ties an entire dictionary component to its corresponding loading vector. Optimizing that scale yields a group-sparsity penalty, and a component is pruned when both sides approach zero. The VEB model above differs in two ways: F is constrained to have unit row sums, and (α_k^0, β_k^0) are fitted to the loading distribution without a sparsity-inducing hyperprior. Learning a prior is therefore not, by itself, equivalent to ARD.

A closer conjugate ARD extension would fix a common loading shape a_L and introduce a factor-specific rate,

$$L_{ik} \mid \tau_k \sim \text{Gam}(a_L, \tau_k), \quad (25)$$

$$\tau_k \sim \text{Gam}(c_0, d_0). \quad (26)$$

Under a mean-field approximation,

$$q(\tau_k) = \text{Gam}\left(c_0 + Na_L, d_0 + \sum_{i=1}^N \mathbb{E}_q[L_{ik}]\right). \quad (27)$$

A factor with little total loading obtains a large posterior rate $\mathbb{E}_q[\tau_k]$, which then shrinks every L_{ik} in that column. A global spike-and-slab indicator for each loading column would provide exact component exclusion, at the cost of a more involved variational update.

For the first experiment, the unconstrained VEB algorithm should be implemented exactly as Algorithm 1, because its successes and failures are scientifically informative. It should then be compared with (i) fixed α_k^0 and learned β_k^0 , and (ii) the hierarchical relevance-rate model in Eq. (27).

9 Proposed validation experiments

For data generated with true rank one, fit $K_{\max} \in \{1, 2, 5, 10\}$ and use multiple random initializations. Record:

1. $r_k = \alpha_k^0 / \beta_k^0$ and the allocated count share p_k ;
2. the thresholded effective rank over a range of τ ;
3. training and held-out Poisson deviance;
4. pairwise cosine similarities among the learned rows of F ;
5. whether redundant factors are pruned or split the true signal;
6. sensitivity to learning both Gamma parameters versus fixing the shape;
7. sensitivity to initialization and update damping.

Held-out prediction is important because a redundant factor may survive by fitting sampling noise in the same data used to learn F and its prior.

References

- [1] V. Y. F. Tan and C. Févotte. Automatic relevance determination in nonnegative matrix factorization with the β -divergence. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(7):1592–1605, 2013. <https://arxiv.org/abs/1111.6085>.